Mohamed Ouais Hamdi

Engineering Student · Artificial Intelligence & Machine Learning

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First-year engineering student at ESIEE-IT (Paris), specializing in AI. I build machine-learning systems end to end and test them on data they were never trained on. **Looking for an internship in AI / machine learning.**

EDUCATION

Engineering Cycle, 1st year (international exchange) | *ESIEE-IT, Paris* *2026 – 2027*

- Focus on artificial intelligence and machine learning; aiming for the two-year AI specialization

B.Sc. in Computer Science (Software Engineering & Information Systems) | *IIT, Sfax* *2024 – 2027*

- Year 1 average: 15.70/20 · Year 2, semester 1: 17.25/20

EXPERIENCE

Full-Stack Software Engineering Intern | *DualCom Consulting, Beirut (remote)* *Jul – Aug 2026*

- Built Restaurant Pro, a restaurant ordering web app: REST API in Node.js, Express and TypeScript on MongoDB, JWT authentication with admin and client roles
- React + TypeScript interface with cart, order tracking and an admin dashboard

PROJECTS

HistoLens | *PyTorch, ResNet-18, ONNX Runtime, FastAPI* *2026*

- Colorectal cancer tissue classification (9 types), trained on one hospital and tested on 7,180 images from another: **95.7% accuracy** (97.8% on the same hospital)
- H&E stain augmentation from scratch; calibrated confidence and a referral threshold (98.7% on the 90% most confident images)

WattCast | *LightGBM, pandas, conformal prediction, FastAPI* *2026*

- Day-ahead forecast of France's hourly electricity demand, backtested over 7.5 years: **2.18% MAPE vs 2.24%** for RTE's published forecast
- Leakage-tested features (calendar, thermal inertia); 80% prediction intervals made reliable with conformal calibration

ThreatScope | *LightGBM, autoencoder, SHAP, FastAPI* *2026*

- Hybrid intrusion detection catching **56.8% of 17 attack types never seen in training** (33.3% supervised alone) at 3.2% false alarms
- Each alert explained with SHAP, mapped to MITRE ATT&CK, with a response playbook

SentinelVision | *YOLO11, ONNX Runtime, Kalman filter, FastAPI* *2026*

- Real-time detection and tracking of people and vehicles with zone-intrusion and loitering alerts: **25 FPS on a 2-core CPU**, no GPU

DocuMind | *Node.js, Express, Gemini API, BM25* *2026*

- RAG document Q&A with streamed, cited answers; hybrid retrieval (BM25 from scratch + embeddings) evaluated on a 40-question benchmark

Ouais FC (portfolio) | *TypeScript, Three.js, cannon-es, Web Audio* *2026*

- Interactive 3D football game where each goal opens a project; procedural graphics, physics and synthesized sound, EN/FR: medouaishamdi.com

TECHNICAL SKILLS

Machine Learning: PyTorch, scikit-learn, LightGBM, ONNX Runtime, SHAP, pandas, NumPy
Methods: Deep learning, computer vision, time-series forecasting, anomaly detection, RAG, model evaluation and calibration
Programming: Python, TypeScript, JavaScript, C, Java, C#, SQL
Engineering: FastAPI, Node.js / Express, React, Docker, Git, pytest, MongoDB, Three.js
Languages: Arabic (native), English (C1), French (B2)